

WALMART SALES PERFORMANCE DASHBOARD

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Course: B.com (FINTECH AND AI)

Tool Used: Microsoft Power BI Desktop

1. Introduction

For this project, I created an interactive **Walmart Sales Performance Dashboard** using Microsoft Power BI Desktop. The main purpose of this project was to analyse Walmart's weekly sales data and present the information through KPI cards, charts, filters and business insights.

The dashboard provides a clear view of important business measures such as total weekly sales, average weekly sales, transaction count and store count. It also helps analyse sales based on stores, holiday and non-holiday periods, fuel price and time.

The Power BI dashboard makes it easier to identify sales trends, compare store performance and understand important business patterns.

2. Objectives

The main objectives of this project were:

- To analyse Walmart's weekly sales data.
- To calculate important sales KPIs using Power BI.
- To understand weekly and monthly sales trends.
- To compare sales performance across different stores.
- To analyse the impact of holiday periods on sales.
- To study the relationship between fuel price and weekly sales.
- To analyse sales performance over different years and dates.
- To create interactive financial visualizations.
- To use filters for better data analysis.
- To identify key business findings and provide recommendations.

3. Dataset Description

The dataset used for this project is the **Walmart Sales Dataset**. It contains information related to Walmart stores and their weekly sales performance.

The major fields used in the analysis include:

- **Store** – Identifies the Walmart store.
- **Date** – Represents the sales date.
- **Weekly Sales** – Shows the sales generated during a particular week.
- **Holiday Flag** – Indicates whether the week was a holiday or non-holiday period.
- **Fuel Price** – Shows the fuel price during the particular period.

These fields were used to create the KPI cards, charts, filters and business insights in Power BI.

4. Data Preparation

The Walmart dataset was imported into **Microsoft Power BI Desktop** for analysis.

The main data preparation steps included:

- Importing the Walmart sales dataset into Power BI.
- Checking the available columns and fields.
- Verifying the data types.
- Checking date and numerical fields.
- Ensuring that the weekly sales values were correctly recognized as numerical data.
- Creating the required calculations and measures.
- Checking the calculations before creating visualizations.
- Organizing the visuals into separate report pages.

Proper data preparation helped ensure that the dashboard presented meaningful and reliable information.

5. KPI Cards

A separate **KPI Cards** page was created to display the major business performance indicators.

The dashboard contains four important KPI cards:

KPI	Value
Total Weekly Sales	6.73 Billion
Average Weekly Sales	1.05 Million
Transaction Count	6K
Store Count	45

Total Weekly Sales – 6.73 Billion

This KPI represents the overall weekly sales generated across the Walmart stores included in the dataset.

Average Weekly Sales – 1.05 Million

This represents the average weekly sales value and provides an indication of the typical sales performance.

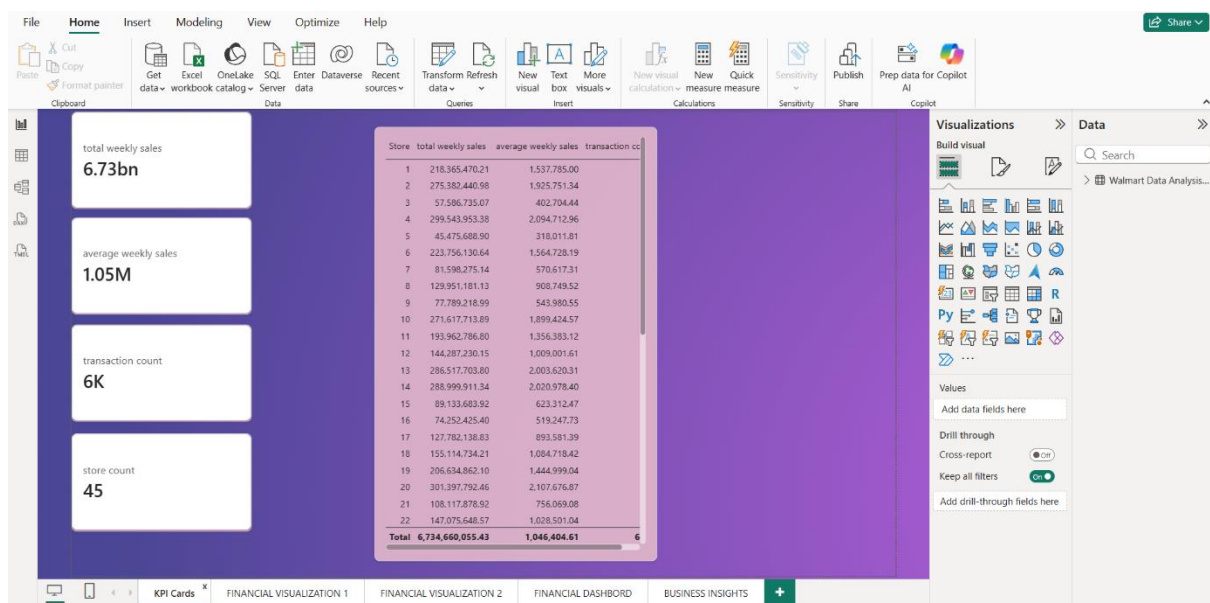
Transaction Count – 6K

This KPI shows the total number of transactions recorded in the dataset.

Store Count – 45

This indicates that the analysis covers **45 Walmart stores**.

These KPI cards provide a quick summary of the overall business performance.



6. Financial Visualization 1

The **Financial Visualization 1** page contains multiple visualizations that provide a detailed view of Walmart's sales performance.

6.1 Total Weekly Sales by Holiday Flag

A **Donut Chart** was used to compare total weekly sales during holiday and non-holiday periods.

The visualization shows:

- Holiday sales: approximately **0.51 billion (7.5%)**
- Non-holiday sales: approximately **6.23 billion (92.5%)**
- Overall sales: approximately **6.73 billion**

The chart indicates that the majority of Walmart's weekly sales were generated during **non-holiday periods**, while holiday periods contributed a smaller share of the total sales.

6.2 Total Weekly Sales by Store

A **Bar Chart** was used to compare weekly sales across the 45 stores.

The chart shows that sales performance varies considerably between stores. Some stores generated significantly higher weekly sales, while others recorded comparatively lower sales.

This helps management identify high-performing and low-performing stores.

6.3 Total Weekly Sales by Year and Store

A **Line Chart** was used to analyse total weekly sales across different years and stores.

The visualization covers the years **2010, 2011 and 2012**. It allows comparison of sales performance across different stores over time.

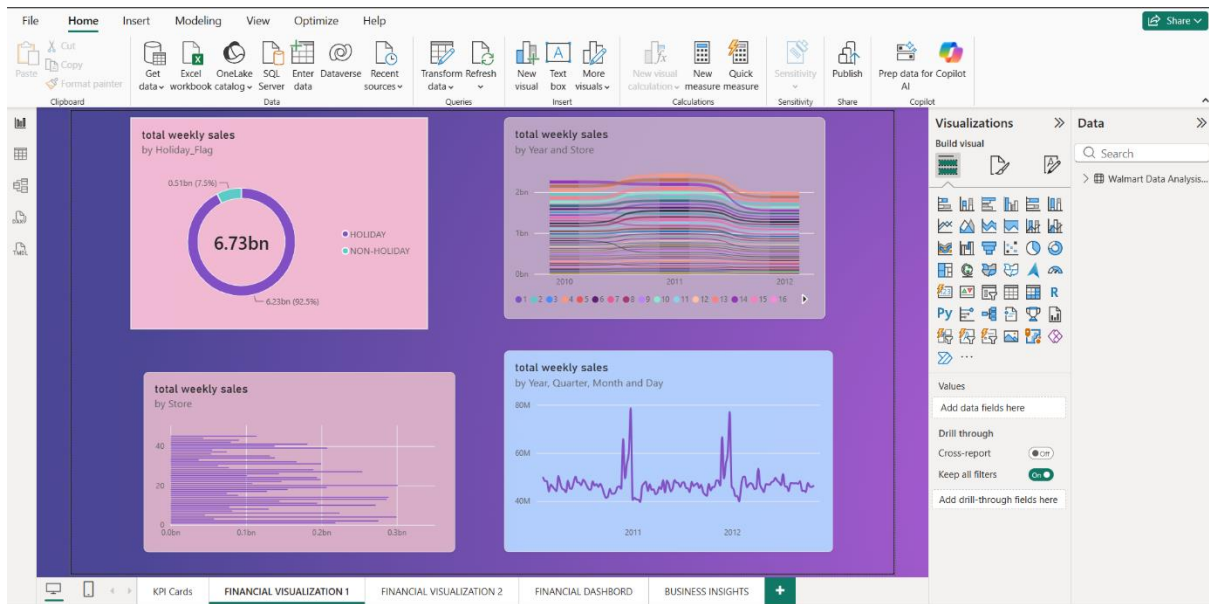
This helps identify changes and patterns in store-level sales performance.

6.4 Total Weekly Sales by Year, Quarter, Month and Day

A **Line Chart** was used to analyse sales across different time periods.

The chart shows weekly sales fluctuations and noticeable spikes during certain periods. The highest peaks are visible around parts of **2011 and 2012**.

This visualization helps identify periods of unusually high or low sales.



7. Financial Visualization 2

The second financial visualization page provides additional analysis using fuel price, time and store information.

7.1 Total Weekly Sales by Fuel Price

A **Scatter Chart** was used to analyse the relationship between fuel price and total weekly sales.

The chart shows that weekly sales vary at different fuel-price levels. However, the points are widely distributed, suggesting that fuel price alone does not completely explain changes in weekly sales.

Therefore, other factors such as store performance, seasonality and holiday periods may also influence sales.

7.2 Average Weekly Sales by Year, Quarter, Month and Day

A **Line/Area Chart** was used to show average weekly sales over time.

The chart shows that average weekly sales generally remain around the **1 million** level, with noticeable increases during certain periods.

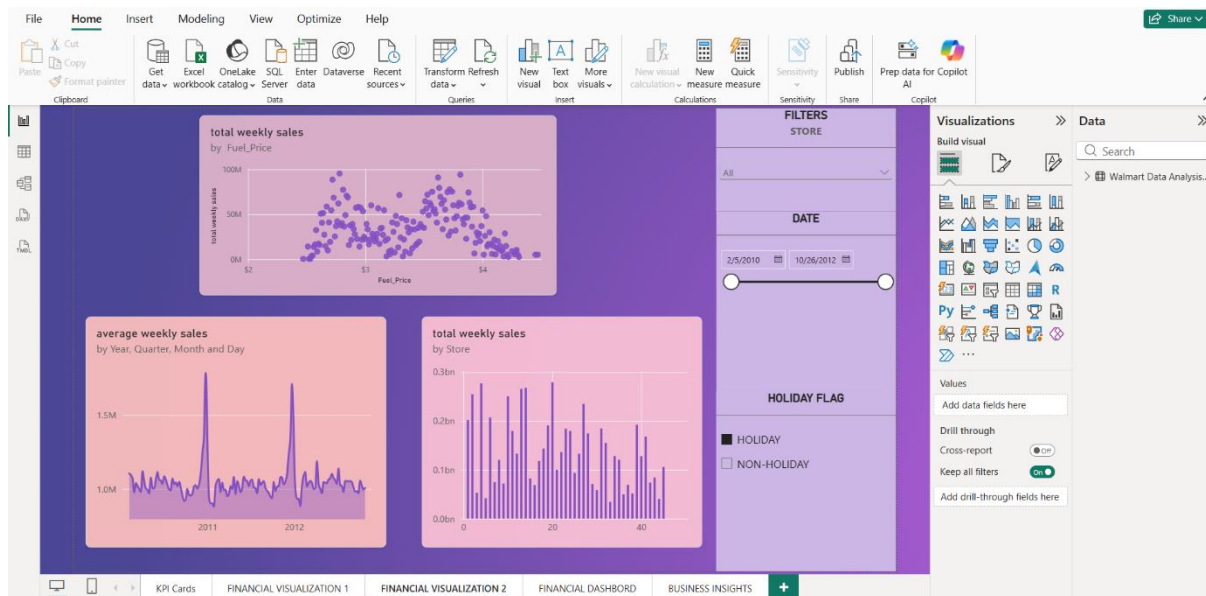
This helps identify periods when average sales were significantly higher than normal.

7.3 Total Weekly Sales by Store

A **Column Chart** was used to compare sales across the 45 stores.

The visualization clearly shows differences in sales performance between stores. Some stores recorded sales close to **0.3 billion**, while others recorded substantially lower sales.

This comparison is useful for identifying stores that require additional attention or improvement.



8. Dashboard Filters

Interactive filters were added to make the dashboard easier to analyse.

The main filters shown in the dashboard are:

Store Filter

The **Store** filter allows the user to select a particular Walmart store and analyse its performance.

Date Filter

The **Date** filter allows users to select a specific period for analysis. The dashboard covers dates from **February 5, 2010 to October 26, 2012**.

Holiday Flag Filter

The **Holiday Flag** filter allows users to compare:

- Holiday
- Non-Holiday

This makes it possible to analyse sales performance during different types of periods.

These filters make the dashboard interactive and allow users to perform customized analysis.

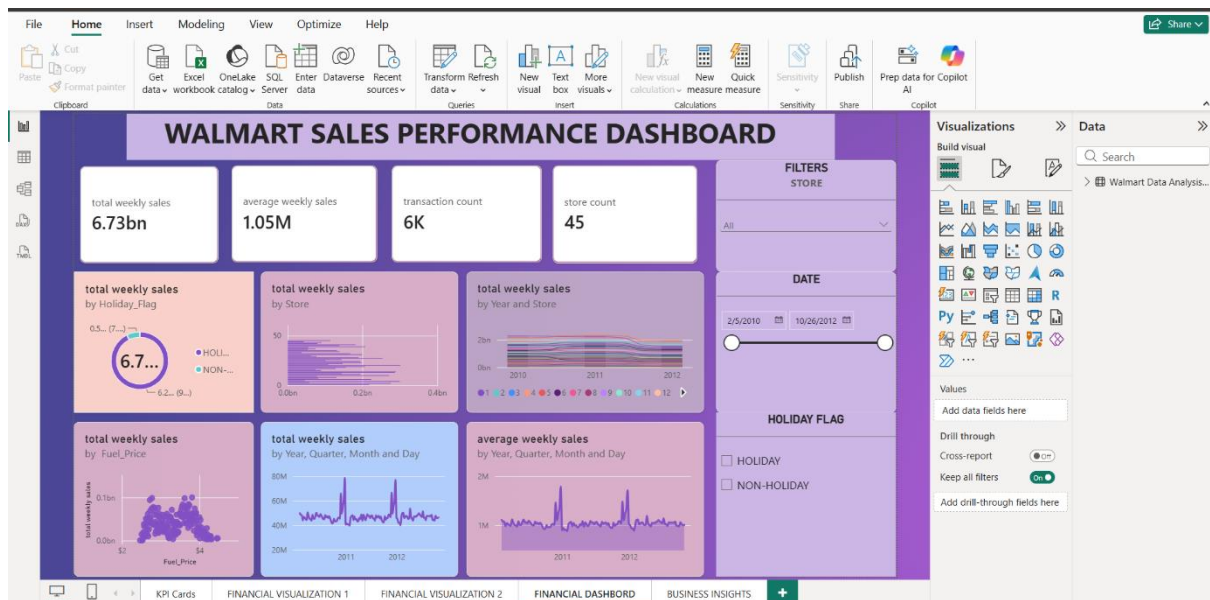
9. Financial Dashboard

The **Financial Dashboard** combines the important KPIs and financial visualizations into a single interactive dashboard.

The dashboard includes:

- Total Weekly Sales
- Average Weekly Sales
- Transaction Count
- Store Count
- Sales by Holiday Flag
- Sales by Store
- Sales by Year and Store
- Sales by Fuel Price
- Sales Trend by Date
- Store and Date Filters
- Holiday Flag Filter

The dashboard provides an overall view of Walmart's financial performance and allows users to analyse different aspects of the business.



10. Business Insights

The **Business Insights** page was created to summarize the major findings from the dashboard and provide suitable recommendations.

Key Findings

1. Store 20 recorded the highest weekly sales.

Among the stores analysed, Store 20 recorded the highest weekly sales performance.

2. Holiday sales contributed a smaller share of total sales.

The dashboard shows that non-holiday sales contributed approximately **92.5%**, while holiday sales contributed approximately **7.5%** of total weekly sales.

3. Fuel price showed limited influence on weekly sales.

The scatter chart indicates that sales were distributed across different fuel-price levels. This suggests that fuel price alone did not strongly determine weekly sales.

4. Certain stores consistently outperformed others.

The store comparison chart shows significant differences in sales between stores, indicating that some stores performed much better than others.

5. Store 33 recorded the lowest weekly sales.

Store 33 was identified as the lowest-performing store based on the dashboard analysis.

6. Sales changed significantly over time.

The time-series visualizations show fluctuations in weekly sales, with noticeable spikes during certain periods.

7. Store-level performance varied considerably.

Across all 45 stores, weekly sales ranged from approximately **37.16 million to 301.40 million**.

8. Store 20 contributed a significant share of total weekly sales.

Store 20 recorded approximately **301.40 million** in total weekly sales and represented around **4.48%** of total weekly sales.

11. Business Recommendations

Based on the findings from the dashboard, the following recommendations can be made:

1. Increase Inventory Allocation for High-Performing Stores

High-performing stores such as **Store 20** should receive sufficient inventory to meet customer demand. Maintaining adequate stock can help prevent stock-outs and support continued sales growth.

2. Introduce Promotional Programmes for Low-Performing Stores

Stores with lower sales performance, such as **Store 33**, can be supported through targeted promotional campaigns, discounts and local marketing activities.

3. Study the Best Practices of High-Performing Stores

Management can analyse the operations, product mix, customer demand and promotional strategies of high-performing stores and apply suitable practices to lower-performing stores.

4. Monitor Sales Trends Regularly

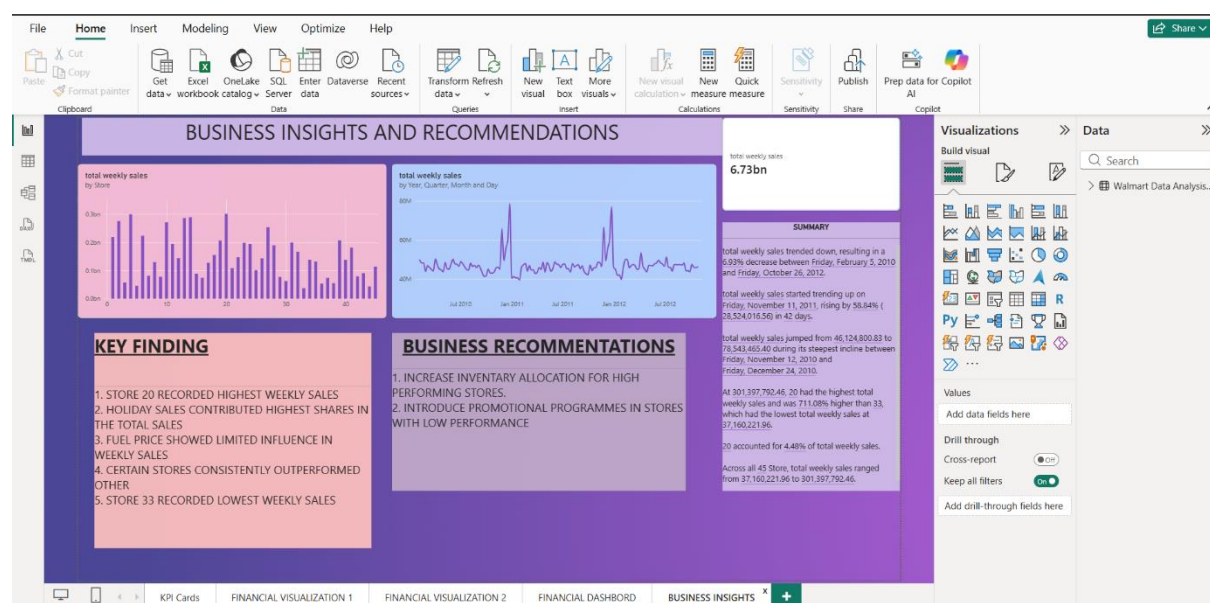
The dashboard should be reviewed regularly to identify sudden increases or decreases in sales. This can help management take timely corrective action.

5. Use Data-Based Store Planning

Store-level data can be used to make better decisions regarding inventory, promotions, staffing and resource allocation.

6. Analyse Holiday Sales Separately

Although holiday sales represent a smaller portion of total sales, special promotional campaigns during holiday periods can help increase customer demand and improve sales.



12. Dashboard Interactivity

The Power BI dashboard was designed to be interactive and user-friendly.

The major interactive features include:

- **Store Filter**
- **Date Filter**

- **Holiday Flag Filter**
- Interactive charts
- Cross-filtering between visuals
- KPI cards
- Time-series analysis

Users can select a particular store, date range or holiday status and observe how the other visuals change accordingly.

This provides a flexible way to analyse the Walmart sales data.

13. Overall Analysis

The Walmart Sales Performance Dashboard shows that the company generated approximately **6.73 billion in total weekly sales**, with an average weekly sales value of approximately **1.05 million**.

The analysis of 45 stores shows that sales performance differs significantly between locations. Store 20 was identified as the strongest performer, while Store 33 recorded the lowest sales.

The holiday analysis indicates that non-holiday periods contributed the majority of total sales. The time-series analysis also shows that sales fluctuate over time, with certain periods recording significant increases.

The fuel-price analysis suggests that fuel price has limited direct influence on weekly sales, meaning that other factors should also be considered when analysing sales performance.

14. Conclusion

This project helped me understand how **Microsoft Power BI** can be used to transform raw sales data into meaningful business insights.

Through this project, I learned how to create KPI cards, financial visualizations, interactive filters and business insights. The dashboard helped me analyse Walmart's total sales, average sales, transactions, store performance, holiday sales, fuel prices and sales trends over time.

The analysis showed that **Store 20 was the highest-performing store**, while **Store 33 recorded the lowest weekly sales**. It also showed that non-holiday periods contributed the majority of total sales and that fuel price had limited influence on weekly sales.

Overall, this project improved my practical knowledge of **Power BI, data visualization, financial analysis, dashboard design and business decision-making**.

15. References

- **Microsoft Power BI Desktop**
- **Walmart Sales Dataset**
- **Microsoft Power BI Documentation**
- Power BI Data Visualization and Business Analytics concepts